

Securing Linux Servers Using Honeypots and IP Blocking

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Date: 06 August 2025

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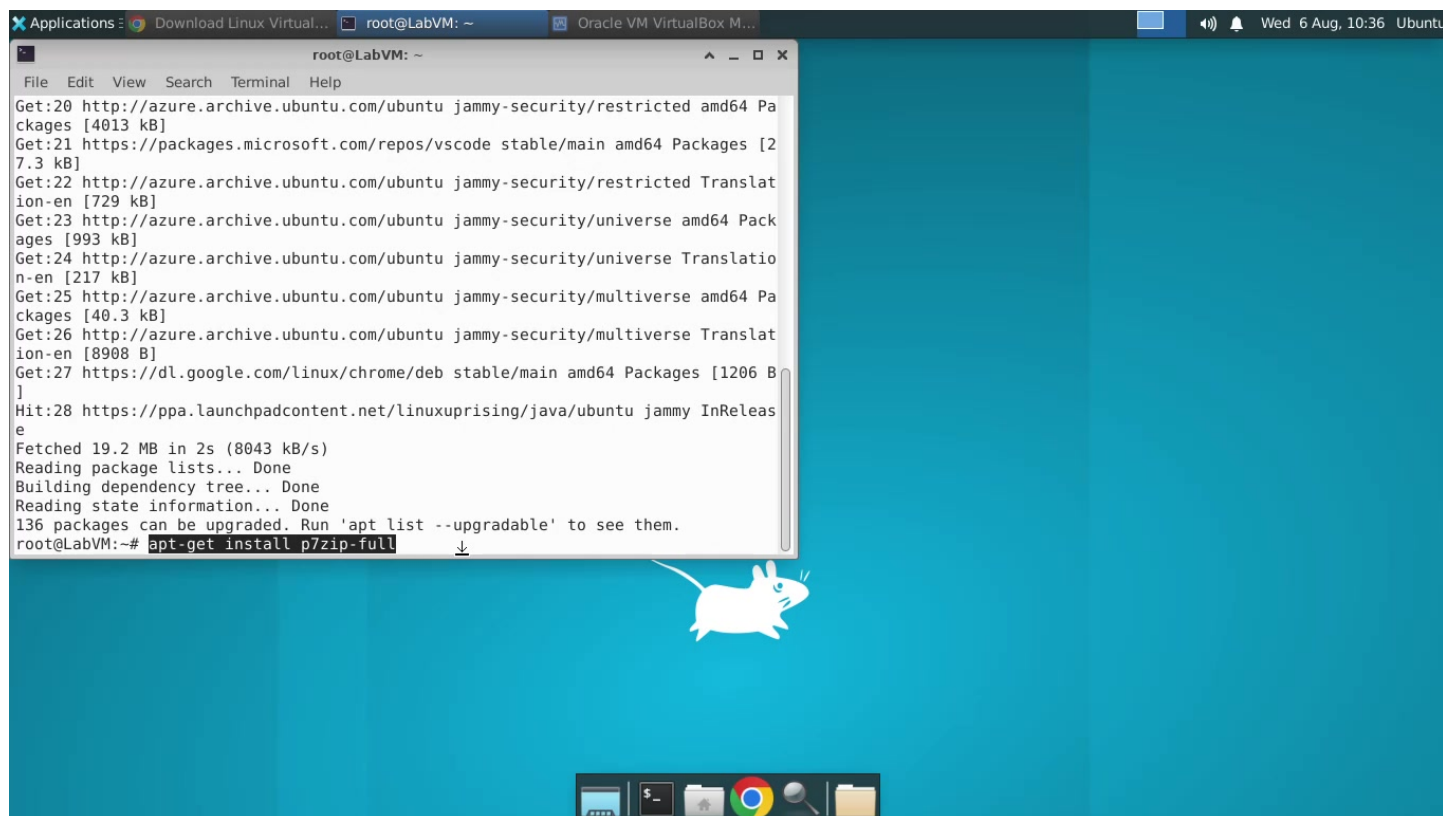
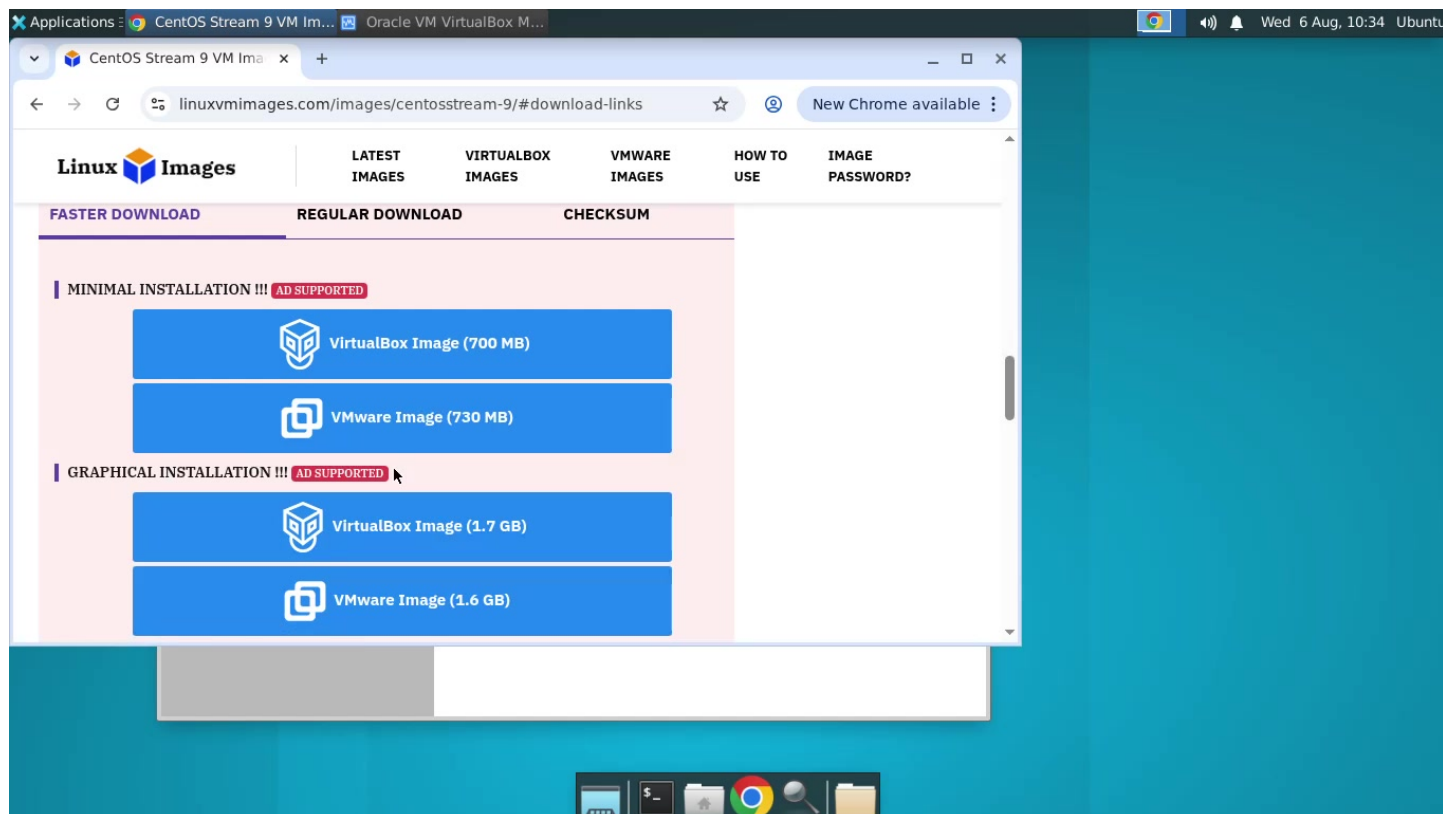
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Introduction

In this project, honeypots and IP blocking techniques were used to secure a Linux server. Honeypots help detect intrusion attempts by attracting attackers, while IP blocking prevents repeat attacks.

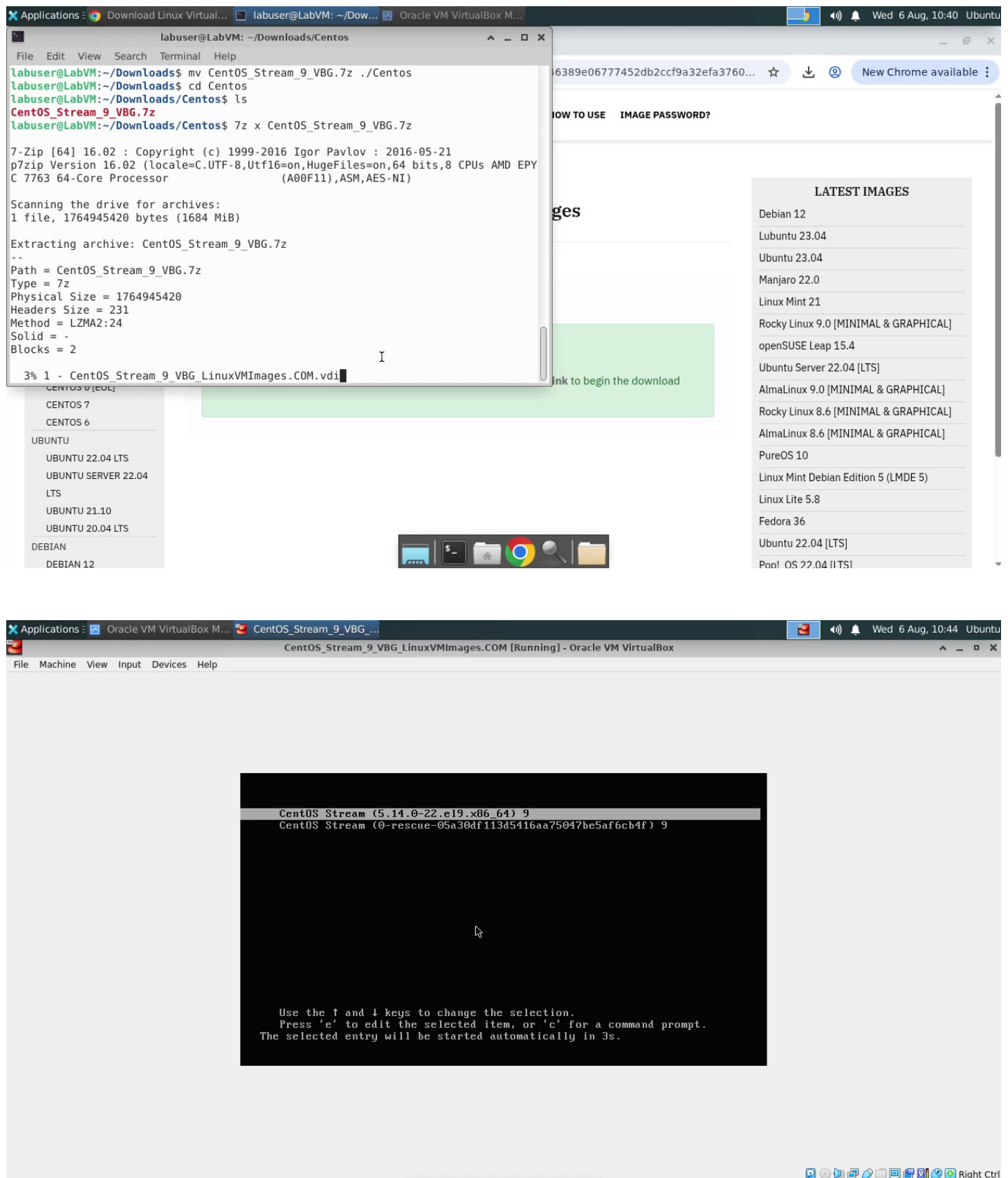
Step 1 - Setting Up the Linux Environment

The Linux server environment was updated and configured to prepare for honeypot deployment.



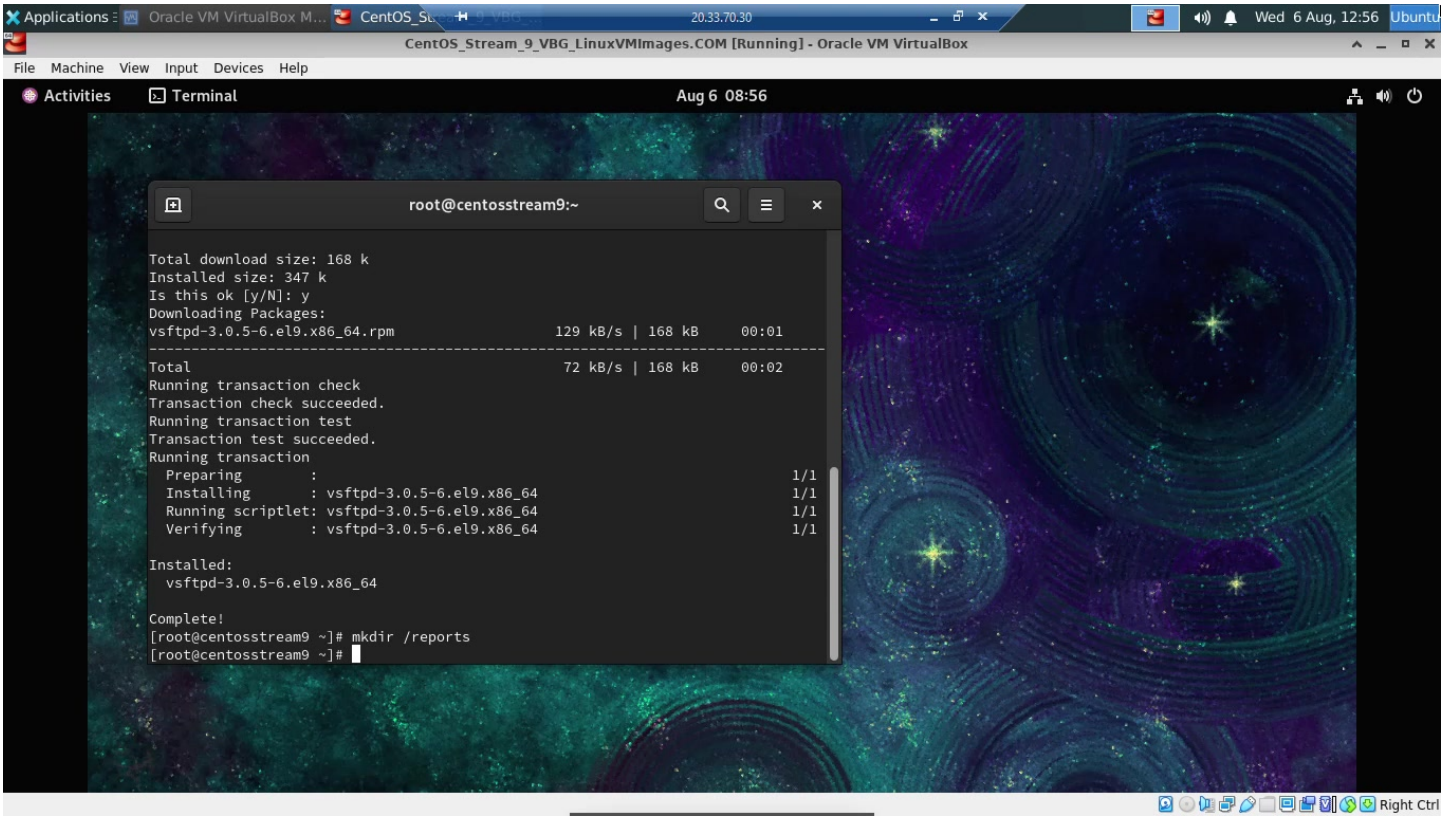
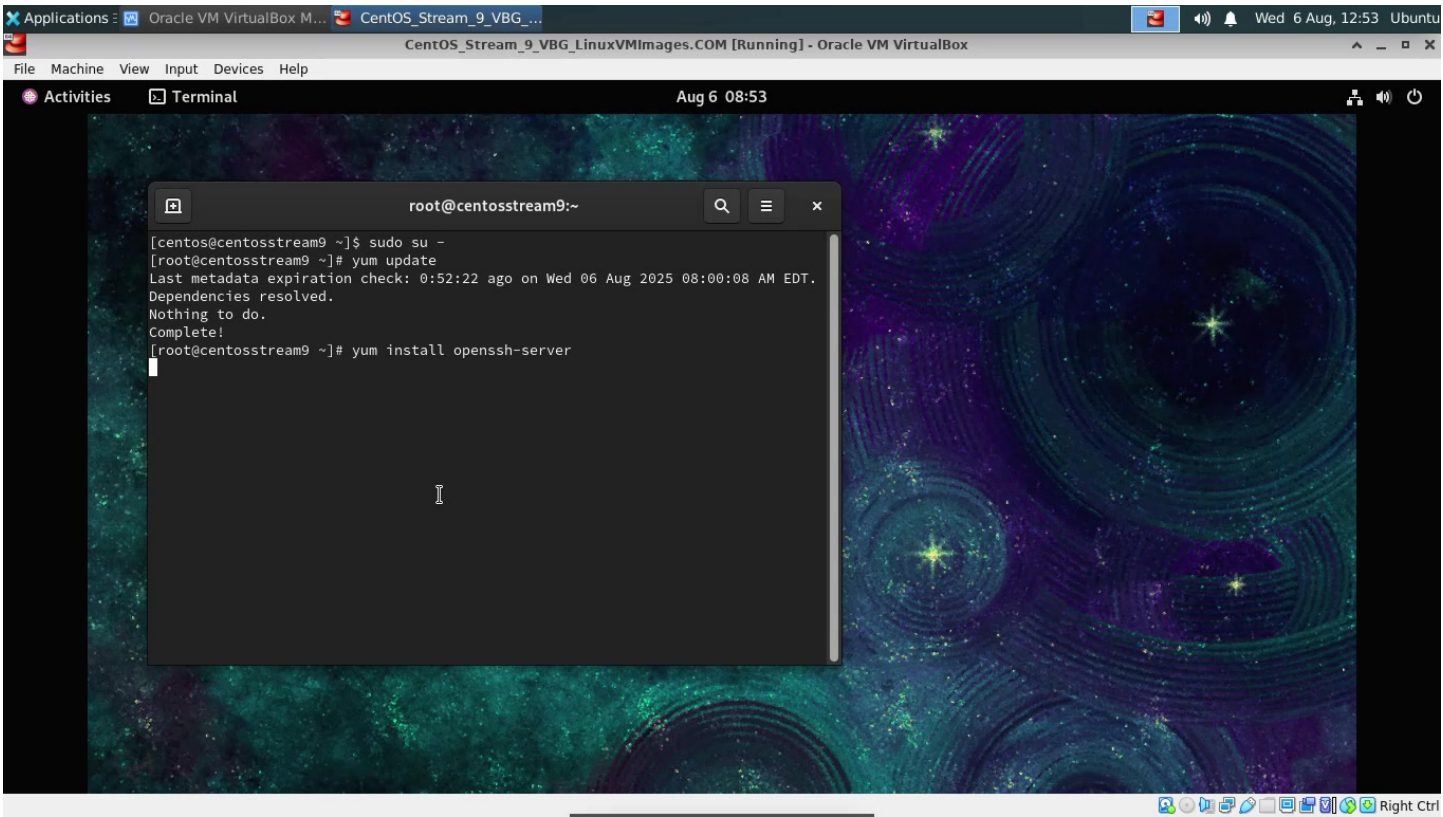
Step 2 - Installing a Honeypot Tool

A honeypot was installed and configured to capture malicious login attempts.



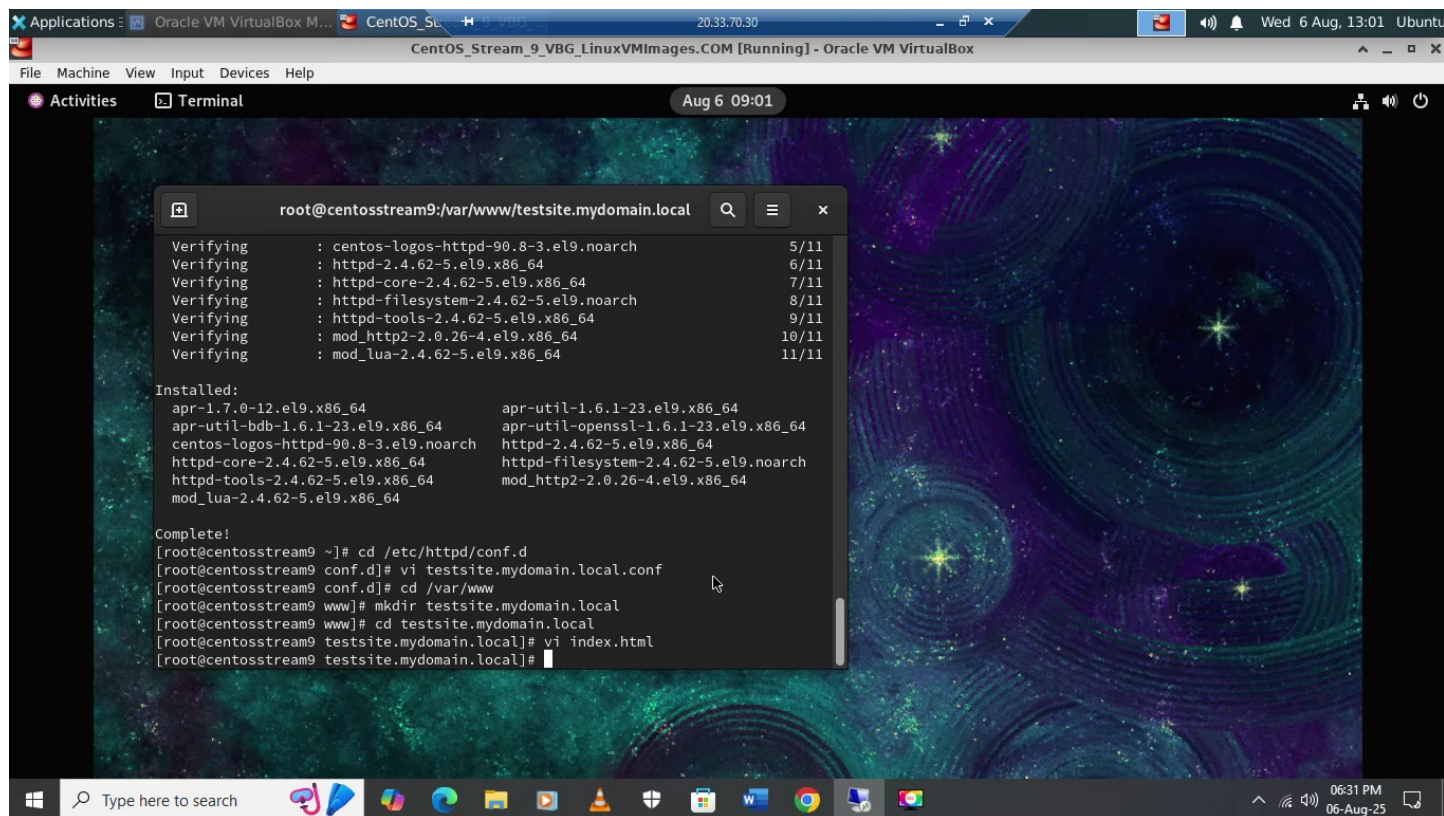
Step 3 - Monitoring Suspicious Activity

The honeypot logs were monitored to track attacker IP addresses and activity.



Step 4 - Implementing IP Blocking

The identified attacker IPs were blocked using firewall rules to prevent further access.



The screenshot shows a terminal window within a VirtualBox environment. The terminal output displays the verification and installation of various packages, followed by the configuration of a web server. The background of the terminal window features a colorful, abstract pattern.

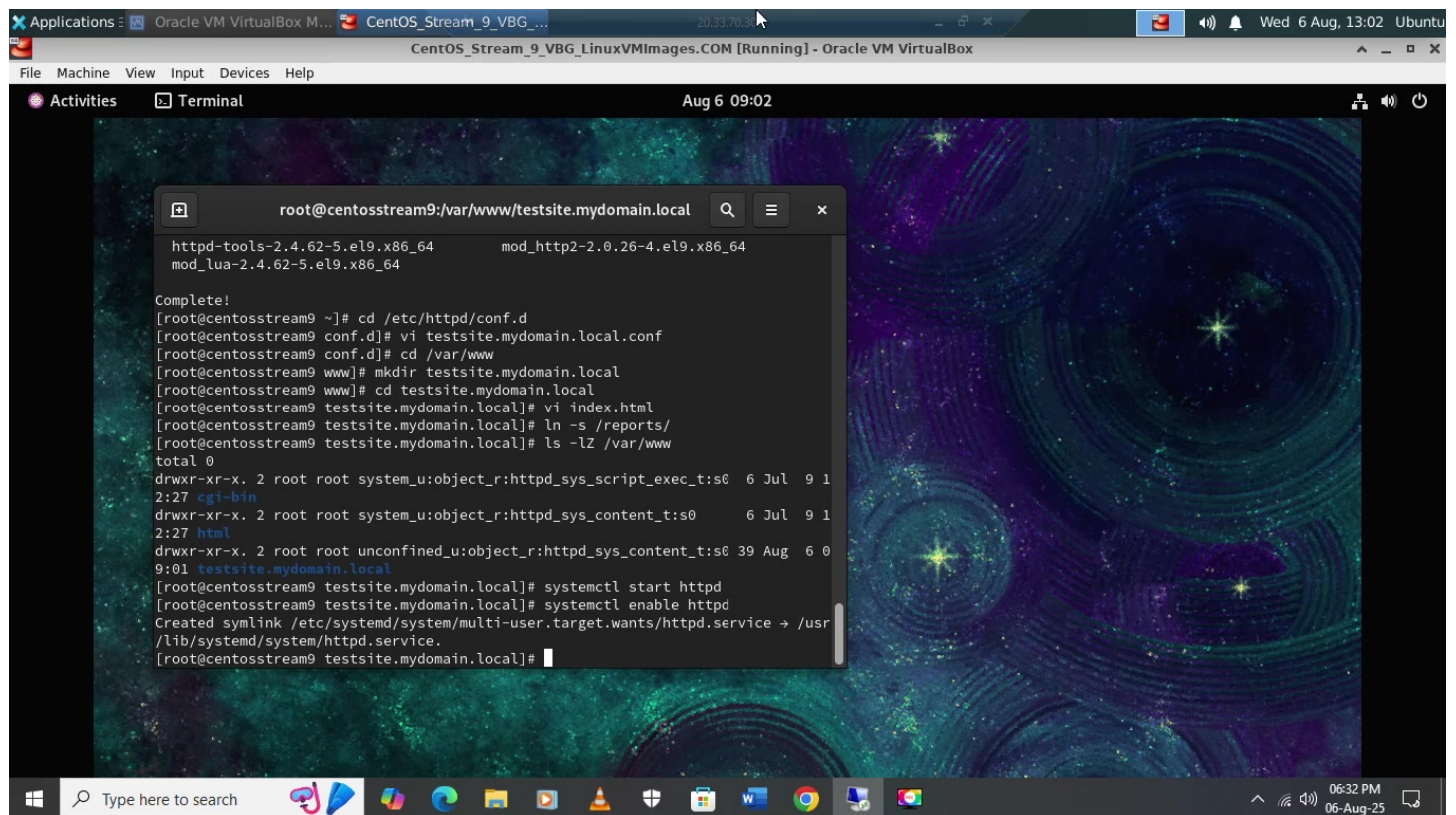
```
root@centosstream9:/var/www/testsite.mydomain.local
Verifying      : centos-logos-httpd-90.8-3.el9.noarch      5/11
Verifying      : httpd-2.4.62-5.el9.x86_64                6/11
Verifying      : httpd-core-2.4.62-5.el9.x86_64           7/11
Verifying      : httpd-filesystem-2.4.62-5.el9.noarch      8/11
Verifying      : httpd-tools-2.4.62-5.el9.x86_64          9/11
Verifying      : mod_http2-2.0.26-4.el9.x86_64            10/11
Verifying      : mod_lua-2.4.62-5.el9.x86_64              11/11

Installed:
apr-1.7.0-12.el9.x86_64      apr-util-1.6.1-23.el9.x86_64
apr-util-bdb-1.6.1-23.el9.x86_64  apr-util-openssl-1.6.1-23.el9.x86_64
centos-logos-httpd-90.8-3.el9.noarch  httpd-2.4.62-5.el9.x86_64
httpd-core-2.4.62-5.el9.x86_64  httpd-filesystem-2.4.62-5.el9.noarch
httpd-tools-2.4.62-5.el9.x86_64  mod_http2-2.0.26-4.el9.x86_64
mod_lua-2.4.62-5.el9.x86_64

Complete!
[root@centosstream9 ~]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]# mkdir testsite.mydomain.local
[root@centosstream9 www]# cd testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]# vi index.html
[root@centosstream9 testsite.mydomain.local]#
```

Step 5 - Automating Defense

Automation was set up to ensure new attacker IPs were automatically blocked.



The screenshot shows a terminal window within a virtual machine environment. The terminal title is 'root@centosstream9:/var/www/testsite.mydomain.local'. The background of the terminal is a colorful, abstract pattern. The terminal output shows the following commands and their results:

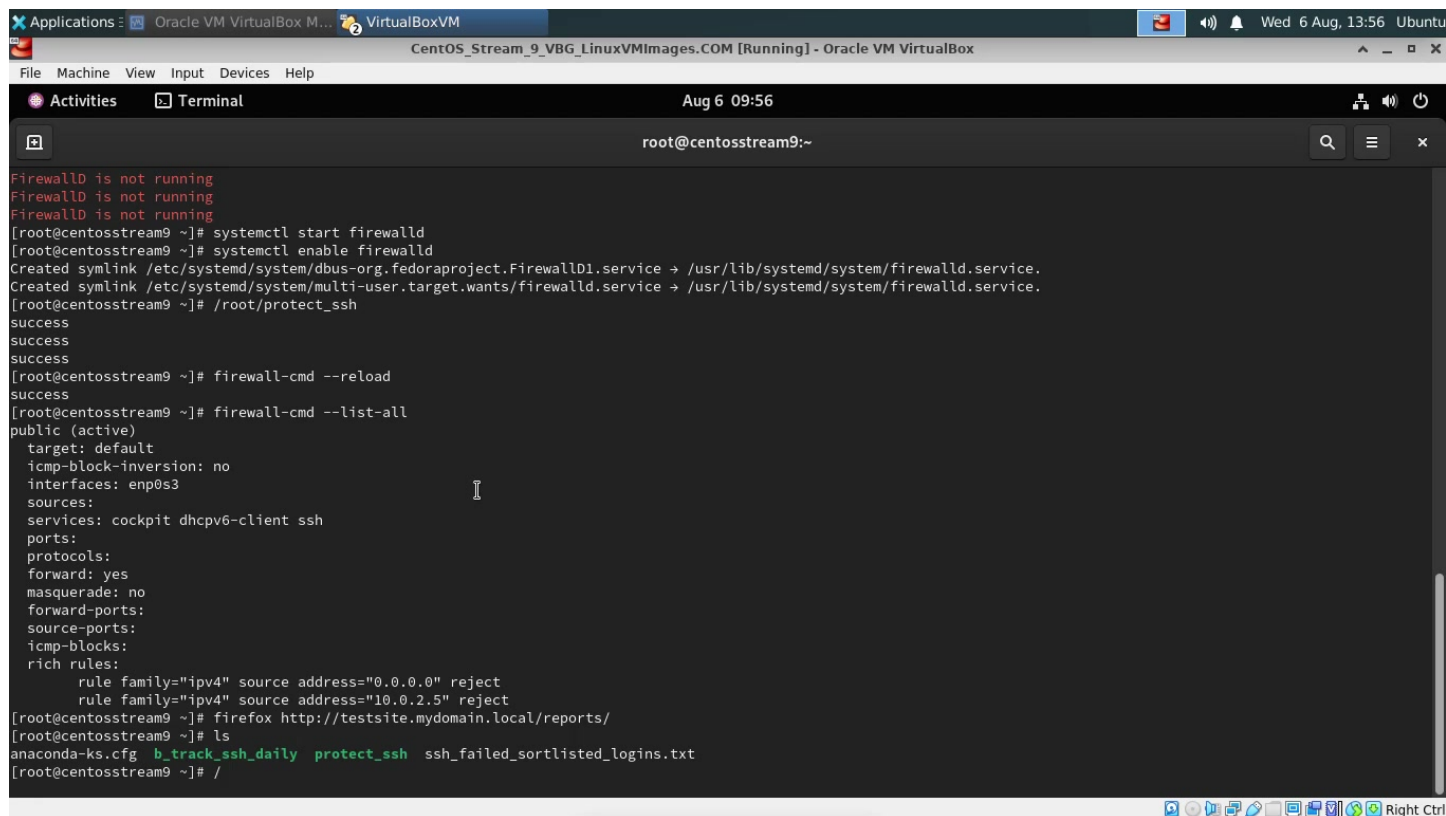
```
httpd-tools-2.4.62-5.el9.x86_64      mod_http2-2.0.26-4.el9.x86_64
mod_lua-2.4.62-5.el9.x86_64

Complete!
[root@centosstream9 ~]# cd /etc/httpd/conf.d
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]# mkdir testsite.mydomain.local
[root@centosstream9 www]# cd testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]# vi index.html
[root@centosstream9 testsite.mydomain.local]# ln -s /reports/
[root@centosstream9 testsite.mydomain.local]# ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0  6 Jul  9 1
2:27  cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0      6 Jul  9 1
2:27  html
drwxr-xr-x. 2 root root unconfined_u:object_r:httpd_sys_content_t:s0 39 Aug  6 0
9:01  testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]# systemctl start httpd
[root@centosstream9 testsite.mydomain.local]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr
/lib/systemd/system/httpd.service.
[root@centosstream9 testsite.mydomain.local]#
```

The terminal window is part of a larger application window titled 'CentOS_Stream_9_VBG_LinuxVMImages.COM [Running] - Oracle VM VirtualBox'. The window has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. The status bar at the bottom shows the date and time as 'Wed 6 Aug, 13:02' and the system as 'Ubuntu'.

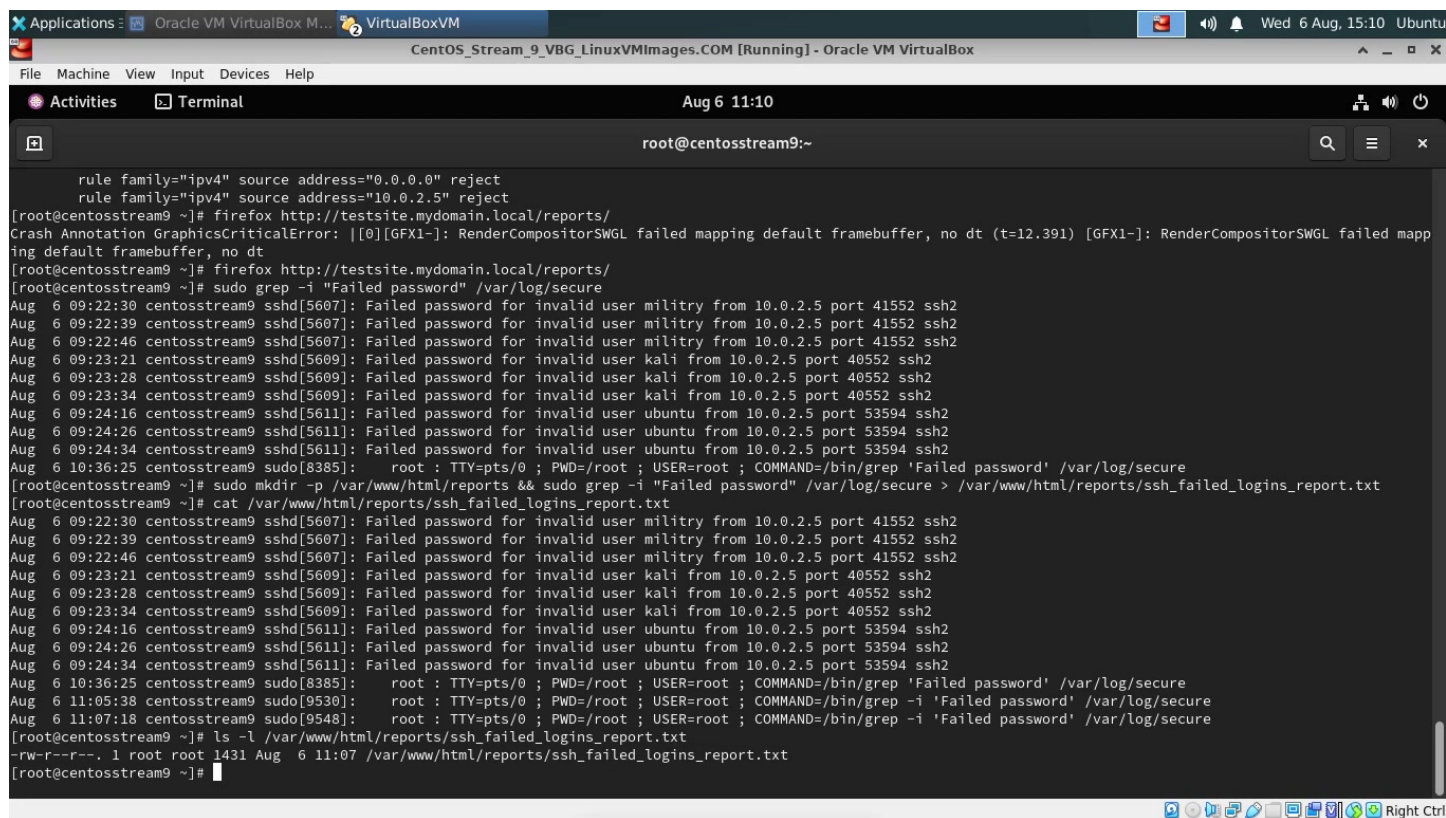
Step 6 - Verification

Verification was done to ensure that the honeypot and firewall were functioning correctly.



```
Applications : Oracle VM VirtualBox M... VirtualBoxVM
CentOS_Stream_9_VBG_LinuxVMImages.COM [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal
Aug 6 09:56
root@centosstream9:~

Firewalld is not running
Firewalld is not running
Firewalld is not running
[root@centosstream9 ~]# systemctl start firewalld
[root@centosstream9 ~]# systemctl enable firewalld
Created symlink /etc/systemd/system/dbus-org.fedoraproject.FirewallD1.service → /usr/lib/systemd/system/firewalld.service.
Created symlink /etc/systemd/system/multi-user.target.wants/firewalld.service → /usr/lib/systemd/system/firewalld.service.
[root@centosstream9 ~]# /root/protect_ssh
success
success
success
[root@centosstream9 ~]# firewall-cmd --reload
success
[root@centosstream9 ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
    rule family="ipv4" source address="0.0.0.0" reject
    rule family="ipv4" source address="10.0.2.5" reject
[root@centosstream9 ~]# firefox http://testsite.mydomain.local/reports/
[root@centosstream9 ~]# ls
anaconda-ks.cfg  b_track_ssh_daily  protect_ssh  ssh_failed_sortlisted_logins.txt
[root@centosstream9 ~]# /
```



```
Applications : Oracle VM VirtualBox M... VirtualBoxVM
CentOS_Stream_9_VBG_LinuxVMImages.COM [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal
Aug 6 11:10
root@centosstream9:~

    rule family="ipv4" source address="0.0.0.0" reject
    rule family="ipv4" source address="10.0.2.5" reject
[root@centosstream9 ~]# firefox http://testsite.mydomain.local/reports/
Crash Annotation GraphicsCriticalError: [0][GFX1-]: RenderCompositorSWGL failed mapping default framebuffer, no dt (t=12.391) [GFX1-]: RenderCompositorSWGL failed mapping default framebuffer, no dt
[root@centosstream9 ~]# firefox http://testsite.mydomain.local/reports/
[root@centosstream9 ~]# sudo grep -i "Failed password" /var/log/secure
Aug  6 09:22:30 centosstream9 sshd[5607]: Failed password for invalid user military from 10.0.2.5 port 41552 ssh2
Aug  6 09:22:39 centosstream9 sshd[5607]: Failed password for invalid user military from 10.0.2.5 port 41552 ssh2
Aug  6 09:22:46 centosstream9 sshd[5607]: Failed password for invalid user military from 10.0.2.5 port 41552 ssh2
Aug  6 09:23:21 centosstream9 sshd[5609]: Failed password for invalid user kali from 10.0.2.5 port 40552 ssh2
Aug  6 09:23:28 centosstream9 sshd[5609]: Failed password for invalid user kali from 10.0.2.5 port 40552 ssh2
Aug  6 09:23:34 centosstream9 sshd[5609]: Failed password for invalid user kali from 10.0.2.5 port 40552 ssh2
Aug  6 09:24:16 centosstream9 sshd[5611]: Failed password for invalid user ubuntu from 10.0.2.5 port 53594 ssh2
Aug  6 09:24:26 centosstream9 sshd[5611]: Failed password for invalid user ubuntu from 10.0.2.5 port 53594 ssh2
Aug  6 09:24:34 centosstream9 sshd[5611]: Failed password for invalid user ubuntu from 10.0.2.5 port 53594 ssh2
Aug  6 10:36:25 centosstream9 sudo[8385]: root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/bin/grep 'Failed password' /var/log/secure
[root@centosstream9 ~]# sudo mkdir -p /var/www/html/reports && sudo grep -i "Failed password" /var/log/secure > /var/www/html/reports/ssh_failed_logins_report.txt
[root@centosstream9 ~]# cat /var/www/html/reports/ssh_failed_logins_report.txt
Aug  6 09:22:30 centosstream9 sshd[5607]: Failed password for invalid user military from 10.0.2.5 port 41552 ssh2
Aug  6 09:22:39 centosstream9 sshd[5607]: Failed password for invalid user military from 10.0.2.5 port 41552 ssh2
Aug  6 09:22:46 centosstream9 sshd[5607]: Failed password for invalid user military from 10.0.2.5 port 41552 ssh2
Aug  6 09:23:21 centosstream9 sshd[5609]: Failed password for invalid user kali from 10.0.2.5 port 40552 ssh2
Aug  6 09:23:28 centosstream9 sshd[5609]: Failed password for invalid user kali from 10.0.2.5 port 40552 ssh2
Aug  6 09:23:34 centosstream9 sshd[5609]: Failed password for invalid user kali from 10.0.2.5 port 40552 ssh2
Aug  6 09:24:16 centosstream9 sshd[5611]: Failed password for invalid user ubuntu from 10.0.2.5 port 53594 ssh2
Aug  6 09:24:26 centosstream9 sshd[5611]: Failed password for invalid user ubuntu from 10.0.2.5 port 53594 ssh2
Aug  6 09:24:34 centosstream9 sshd[5611]: Failed password for invalid user ubuntu from 10.0.2.5 port 53594 ssh2
Aug  6 10:36:25 centosstream9 sudo[8385]: root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/bin/grep 'Failed password' /var/log/secure
Aug  6 11:05:38 centosstream9 sudo[9530]: root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/bin/grep -i 'Failed password' /var/log/secure
Aug  6 11:07:18 centosstream9 sudo[9548]: root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/bin/grep -i 'Failed password' /var/log/secure
[root@centosstream9 ~]# ls -l /var/www/html/reports/ssh_failed_logins_report.txt
-rw-r--r-- 1 root root 1431 Aug  6 11:07 /var/www/html/reports/ssh_failed_logins_report.txt
[root@centosstream9 ~]#
```

Summary

The project successfully demonstrated how honeypots combined with IP blocking can effectively enhance Linux server security.